



Leslie Willcocks

June 7th, 2025

Is Starbucks' reversal of automation the new game in town?

0 comments | 7 shares

Estimated reading time: 7 minutes



*Starbucks thought it was streamlining drink-making and reducing labour costs when it introduced an automated system in 2022. Three years and many disappointed customers later, the system is being rolled back. **Leslie Willcocks** analyses what makes technology rollouts work and what often goes wrong in the process.*

In April 2025, Starbucks announced it was pulling back on its **Siren System** rollout begun in 2022 and designed to streamline drink-making and labour costs. Against lower-than-expected US sales and shrinking profit margins, recently appointed CEO Brian Nicholl would reverse replacing labour with machines and, despite additional costs, quickly bring back more baristas into its 3000 US stores. Three months earlier, Chipotle, a major competitor, had announced major new investments in technology. And nimble start-up Seven Brew Coffee spotted the trend towards drive-thru coffee kiosks and made competition there both challenging and necessary. What to think?

Well, headlines intimate a story that can be misleading. For example, far from being a naïve user of digital technologies, Starbucks has been a pioneer, and in a digital transformation journey of some eight years so far, deploys multiple suites of systems very successfully. However, in the very competitive drinks and food service sector, many customers had become more cautious financially; some reported the “feel” at Starbucks had become more transactional and less about the customer experience, while, for others, Starbucks’ recent labour strife detracted from employee satisfaction and, more broadly, what the brand stood for.

Starbucks also operate in-café, mobile order and drive-thru options, and there are challenges customising technology for all three channels. Mobile ordering and drive-thru have been seen as

solutions to slumping sales, but mobile ordering has led to overcrowding and queues in cafes, and drive-thru a greater focus on economy and speed at the expense of customer engagement.

Where then does technology figure in all this? Major technological investments have to function in a system of work that attempts to align and optimise strategy, management attitudes, processes, people, culture, legacy data and technologies, and organisation structure. In such settings, we consistently find that roughly 70 per cent of the challenges are managerial and organisational, and 30 per cent have to do with the technology. Technology can surely help. As a small example, CEO Nicholl observes that the mobile ordering algorithm needs to be adjusted to reduce the over-queuing in café environments. Too often technology is designed from an internal efficiency rather than customer experience perspective. Starbucks is now setting about adjusting their own imbalances on this.

Over the years we have identified lasting management principles that need to be applied to turbulent technology. The Starbucks case suggests six major ones:

Competing on service with technology

In service industries, what competes is what differentiates at the point of customer experience. Put bluntly, service quality matters. In the well-known SERVQUAL model, the best predictors of customer delight are doing well on five dimensions —tangibles, reliability, responsiveness, assurance, and empathy. On tangibles, one factor is that many customers think the coffee price is too high, for the

service and ambience they experience. Starbucks' response seems to be to focus on enhancing the experience rather than reducing the price – will this work?

Customers also experience the café environments as more stripped down and not like pre-Covid, something that Starbucks is now remedying. Tangibles include the taste of the coffee, which according to Neapolitan coffee lore is dependent on the “Four Ms” – miscela, macinacaffe, machina and mano – which we will translate as mix, coffee grinder, coffee machine and barista. That is, machines can be at best only 50 per cent of the story. For example, automation can do quite a lot on reliability and speed up processes (responsiveness), but it is the staff that gives you assurance and empathy.

The amplifier effect of technology

Our research has consistently found over many years that a good management is made better, often much better by technology, but a poor management much worse. If you apply technology to a work system that has poor components or is not properly aligned, you are more likely to get what we call “disaster faster”. In their **research**, Stuart Mills and David Spencer found a recent example, recording that when AI is applied to “bullshit jobs” the result is “efficient inefficiency”.

At Starbucks, it may be that the complexity of the drink products, innovation and work requirements on staff meant that the technology was being applied prematurely to a work system that first needed more simplification. Without this, perhaps Starbucks eroded some of its points of difference in the customer experience.

Management visions matter

Digital technologies can be used for a variety of purposes. Most typically we have come across four executive visions: to automate, mainly to displace labour and achieve efficiency; to “**informate**” up to allow senior management to control and micro-manage; to informate down to give more information to and empower those working in lower parts of the organisation, including in customer-facing roles; and/or to transform the organisation and its relationships with customers, suppliers, competitors, and/or complementors. In 2025, Starbucks would seem to have several visions operating, but were leaning too much on the automation and information up visions with a number of important technology applications.

The technology navigation dilemma

Computing and digital technologies have always been massively hyped. This seems to have only peaked these days with AI. A tool we have deployed for many years to help executives find their way suggests that there are four hi-tech domains to navigate.

The first is **technology hype**. This is the domain and agenda of technology vendors, management consultants and the media. In practice all media like to polarise technology stories into either examples of massive success or disaster, autotopia or automageddon, as it were. The second domain is **technology capability**. This is the domain of technologists and those selling hardware,

software and applications. Too often it turns out that the technology is a solution looking for a business problem it can solve. A lot of recent AI is probably at that stage.

The third domain is what I will call “**useful**” **technology**. This is an even more dangerous domain because even a child of six years of age can make a persuasive case for more technology. Think what happens in a large business enterprise dealing in billions of dollars where powerful people can make claims for resources, and organisational politics is alive and well. The fourth domain is **strategic technology**, that is, the relatively few technologies/systems/applications that will make a disproportionate contribution to business value and strategic direction.

Of the four agendas only the fourth – strategic one – should be the one adopted as the business agenda, as opposed to the hype, capability and user agendas. In Starbucks’s case I suspect the Siren System was identified as strategic across the business, but now a more limited role has been correctly identified.

The implementation challenge

Unfortunately, a technology is rarely a silver bullet or a “fire -and-forget missile”. To achieve business value they have to be designed, developed, trialled, adopted and then institutionalised, meeting many challenges, with many adjustments being made on the long road to effectiveness. It can take years to bed a technology down to achieve its optimal business value. In competitive environments, the less useful ones get abandoned along the way. In the Starbucks case, the Siren system had been

implemented in about 10 per cent of US outlets, and the plan had become to install the system only in very targeted stores, such as those with high drive-through customer volumes and overall sales.

The digital leadership quest

Depending on sector, between 15 and 23 per cent of medium to large organisations will be digital high performers. Our most **recent research** shows that another 25 per cent will be followers, 35 per cent laggards, and another 20 per cent will be “nascent” digital organisations. On my reading of the secondary literature, Starbucks in the food and drink sector may be a digital leader. This requires being very good at seven core capabilities, namely strategy, integrated planning, embedded culture, program governance, digital platform, change management and navigation capabilities.

Note that, of these, technology forms really only one core capability. The rest are management core capabilities. Core capabilities differentiate an organisation competitively by being rare, valuable, not easily replicated or substituted and arise as distinctive and required creations of that organisation.

An interesting question is: how long can a business like Starbucks hold on to such a digital leadership position? One worrying signal is when capabilities fail to translate into superior business results. In some sectors, we have found digital leaders widening the gap between them and the rest, so as to make their competitive advantage almost irreversible. In others they falter in important areas, and followers can catch up quickly. Shaping advanced technologies for competitive advantage is a key area, but, as in the case of Starbucks, technology is not the all-consuming answer or disaster that headlines too often like to suggest. In a customer service business, the small details

are a point of difference, the customer experience is paramount, and it is these, and the Starbucks CEO's determination to reintroduce what the brand is supposed to stand for that the technology will need to underpin.

- *Many of the author's insights are shared in his latest book [Maximizing Value With Automation and Digital Transformation A Realist's Guide](#) (Palgrave, 2024) and a forthcoming one, [Automation Anxiety: From Computing to AI](#) (2025).*
 - *This article first appeared at the [LSE Business Review blog](#).*
 - *Subscribe to [LSE USAPP's email newsletter](#) to receive a weekly article roundup.*
 - *Featured [image](#) provided by Shutterstock.*
 - *[Please read our comments policy before commenting.](#)*
 - *Note: This article gives the views of the author, and not the position of USAPP – American Politics and Policy, nor the London School of Economics.*
-

About the author



Leslie Willcocks